Small RTL mistakes, explained

Day 5 questions and submission notes: operators, concatenation width, sticky edge capture and one-hot Mealy encoding.

Kapil Tripathi | Verilog & digital design | 12 pages

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KEEP THIS IDEA

Use one counterexample to distinguish similar-looking expressions. Keep the earlier attempt beside its correction.

Tracker entries: 81, 82, 83, 84, 85

Study notes by Kapil Tripathi, based on HDLBits exercises. Original questions, source references and dated verification records are retained in the notes.

ALL NOTES + ENTRY INDEX HDLBits problem collection

Day 5 questions and submission notes

This review explains the saved code for entries 81-85 and keeps the earlier questions for entries 54, 74, 75, 77 and 80 nearby. The submission records at the end were collected on 2-3 September 2026. They describe that checkpoint; the tracker holds the current progress.

Entries 81-85 are all Done in the tracker. The main fixes are Boolean OR, the two padding bits in a concatenation, sticky edge capture, the ring/vibrate condition, and one-hot Mealy state encoding. Entry 80 has its own detailed Moore explanation.

This PDF contains the review and its historical submission table. The Markdown companion remains available from the collection index and the existing tracker links.

The saved records show success through entry 93 at the original checkpoint. Entries 86-93 were confirmed by Kapil on 3 September. A matching tab alone does not show that its current editor contents pass; later experiments can differ from an earlier successful submission.

Earlier questions and where to read them

Entry	Exact subject found in submitted code	Existing answer
7	Why the Lemmings4 fall counter does not need another reset	Combined questions PDF, page 8
13	Why bitwise operators are used in the vector mux	Combined questions PDF, section 6
26	Why `initial` cannot initialize a live population count and when to use procedural versus generate loops	Combined questions PDF, sections 2, 3, and 5
29	How `sel[0]` and `sel[1]` divide work between the two mux levels	Combined questions PDF, section 6
30	Whether the 101-node carry chain should be a wire or register	Combined questions PDF, section 4
31	Difference between bitwise `~` and logical `!`	Combined questions PDF, section 9
36	Why `&` and `&&` happen to choose the same branch with an all-ones mask	Combined questions PDF, section 10
38	What latch inference means for an intentionally incomplete assignment	Combined questions PDF, section 11
67	Meaning and tokenization of indexed part-select operators `+:` and `-:`	Day 4 questions PDF, pages 2-3
70	Why the PS/2 data register must not be cleared in an unmatched clocked `else` branch	Day 4 questions PDF, PS/2 datapath section

Question 1 - Entry 54, water-level Moore FSM

Original code comment:

```
//localparam s2 // i dont need the check for specific cases cause they are already in the input
```

The input tells the controller which water-level sensors are currently asserted, but it does not necessarily contain all the history needed by the outputs. In this problem, the ordinary flow outputs are determined by the current level band, while the supplemental-flow decision also distinguishes whether the level arrived from above or below. Two moments can therefore have the same current sensor pattern but require different `dfr` behavior because their preceding level was different.

`localparam` declarations do not test inputs. They give readable binary encodings to internal states. States such as `belows2` and `aboves2` preserve the direction/history that the input vector alone cannot express. The combinational next-state block must still test the valid sensor bands to decide which history state comes next.

If the HDLBits statement guarantees physically valid sensor patterns, it is reasonable to prioritize the highest asserted sensor rather than enumerate impossible patterns separately. That does not remove the need for the history states. This fragment shows the idea (it is not a complete submission):

```
always @(*) begin
    next_state = state;
    if (s[3])
        next_state = ABOVE_S3;
    else if (s[2])
        next_state = (state_was_higher) ? BELOW_S3 : ABOVE_S2;
    else if (s[1])
        next_state = (state_was_higher) ? BELOW_S2 : ABOVE_S1;
    else
        next_state = BELOW_S1;
end
```

The exact state names may differ, but the principle is fixed: sensor bits describe the present level; state bits preserve any past information the outputs still need.

Question 2 - Entry 74, sampled positive-edge detection

Original code comment:

`// is it like 0 to 1 transition for any clock edge ? like in any 2 clock edges its that, transition`

Yes, but "edge" here means a change between **two consecutive rising-clock samples**, not every physical transition that might occur between clocks. For each vector bit:

- `previous[i] = 0` at the preceding rising edge,

- `in[i] = 1` at the current rising edge,

- therefore `in[i] & ~previous[i] = 1` for one cycle.

The sequential implementation is:

```
reg [7:0] previous;

always @(posedge clk) begin
    previous <= in;
    pedge    <= in & ~previous;
end
```

Both right-hand sides see the old `previous` value because nonblocking assignments update after the block has evaluated. The result is one pulse per sampled 0-to-1 transition. A pulse that rises and falls entirely between two clock edges may be missed because this is synchronous sampling, not an asynchronous edge detector.

The unsuccessful submission used `~in & previous`, which detects the opposite sampled transition: 1-to-0.

Question 3 - Entry 75, asynchronous-reset sensitivity list

Original code comment:

```
// why i have to write, asynchronous reset in the sensitivity list
```

An asynchronous reset must change the state register immediately when reset is asserted, without waiting for a clock edge. The event control must therefore wake the block for either event:

```
always @(posedge clk or negedge aresetn) begin
    if (!aresetn)
        state <= S0;
    else
        state <= next_state;
end
```

`aresetn` is active low, so assertion is its falling edge, written `negedge aresetn`. A bare `aresetn` in the event list does not describe the required flip-flop primitive and does not state which transition asserts reset. Omitting reset from the list would make the reset synchronous because the block could react only at `posedge clk`.

In physical designs, asynchronous assertion is often paired with synchronized deassertion to avoid recovery/removal timing problems, but the HDLBits problem specifically tests the basic active-low asynchronous-reset behavior.

Question 4 - Entry 77, reduction versus bitwise versus logical operators

Original code comment:

```
// tell me difference b/w clearly b/w reduction and logical and bitwise
```

Operator class	Example	Input/output width	Meaning for `in = 4'b1011`
Unary reduction	`&in`, `|in`, `^in`	Vector to one bit	`&in=0`, `|in=1`, `^in=1`
Bitwise binary	`a & b`, `a | b`, `a ^ b`	One result bit per aligned input bit	Combines corresponding bits independently
Logical	`a && b`, `a || b`, `!a`	Operands become true/false; result is one bit	Tests whether each whole operand is zero or nonzero

The original problem asks for one output from all four bits, so unary reduction operators are the direct answer:

```
assign out_and = &in;
assign out_or  = |in;
assign out_xor = ^in;
```

The unsuccessful forms `out_and &= in`, `out_or |= in`, and `out_xor ^= in` are compound read-modify-write assignments. They read the old output and combine it with `in`; they are not reduction operators. In combinational logic, reading an output while assigning that same output can also create an unintended feedback dependency.

Submission notes for entries 80-85

Entry	Chrome evidence	Attempt 2 state	Finding
80	Last success 2026-09-02 16:28:11; last non-success 16:18:12	Done	The corrected Moore implementation passed, and its new in-code explanation request is answered in a dedicated PDF.
81	Last success 2026-09-01 23:26:08	Done	Waveform implements `q = b | c`; inputs `a` and `d` are irrelevant.
82	Last success 2026-09-01 23:23:20	Done	Correct 32-bit concatenation includes two trailing one bits.
83	Last success 2026-09-02 17:00:08; last non-success 16:57:13	Done	The successful version keeps captured falling edges sticky and initializes the previous sample during reset.
84	Last success 2026-09-02 16:54:21; last non-success 16:53:47	Done	The successful motor equation now requires both `ring` and `vibrate_mode`.
85	Last success 2026-09-02 16:44:54; last non-success 16:31:02	Done	The two-state Mealy behavior passed; the internal encoding is still not literally one-hot.

Entry 80 - Q5a serial two's complementer, Moore FSM

For an LSB-first stream, two's complement can be produced by copying zeros until the first 1, copying that first 1, and inverting every later bit. A Moore output cannot depend directly on the current input, so the output value must be represented by the state reached after sampling that input. This needs three functional states:

- `COPY_ZERO`, output 0, before the first 1;

- `OUTPUT_1`, output 1, for the first 1 or a later inverted 0;

- `OUTPUT_0`, output 0, for a later inverted 1.

The saved successful code follows this rule and includes a `default` transition for the unused state encoding. Its submitted comments explicitly ask for a deeper explanation of the algorithm, timing, and need for each state; see the Entry 80 Moore FSM deep-dive PDF.

Entry 81 - Combinational circuit 4

The waveform shows that `q` depends only on `b` and `c`:

```
assign q = b | c;
```

The old `b + c` expression is arithmetic addition, not Boolean OR. When `b=c=1`, the mathematical sum is binary 2 and the carry cannot be represented by the one-bit output, while OR must remain 1. This counterexample distinguishes the two immediately.

Entry 82 - Vector concatenation

Six five-bit inputs contain 30 bits, while `{w,x,y,z}` contains 32 bits. The original prompt explicitly requires two one bits after the six vectors:

```
assign {w, x, y, z} = {a, b, c, d, e, f, 2'b11};
```

Omitting the final two bits does not merely leave `z[1:0]` unspecified. The 30-bit value is extended to the 32-bit destination from the left, so the grouping of all output bytes is shifted relative to the required concatenation.

Entry 83 - Sticky falling-edge capture

The latest non-success overwrote `solution` with only the newest falling-edge mask. The prompt requires every detected bit to remain set until reset, so the new mask must be ORed with the old output. The previous input must also be sampled on every edge, including reset edges, so the first comparison after reset has a defined predecessor. The latest successful submission applies both corrections.

```
reg [31:0] previous;

always @(posedge clk) begin
    previous <= in;
    if (reset)
        out <= 32'b0;
    else
        out <= out | (previous & ~in);
end
```

Reset has priority over capture. `previous & ~in` detects bits that were 1 at the preceding sample and are 0 now. The OR makes the result sticky.

Entry 84 - Ring or vibrate

The required truth table is:

ring	vibrate_mode	ringer	motor
0	0	0	0
0	1	0	0
1	0	1	0
1	1	0	1

Therefore:

```
assign ringer = ring & ~vibrate_mode;
assign motor  = ring &  vibrate_mode;
```

The unsuccessful equation used `vibrate_mode & !ring`, which turns the motor on when no call is arriving and turns it off for the exact `ring=1, vibrate_mode=1` case that should activate it. For one-bit signals, `!ring` and `~ring` have the same 0/1 result, so the failure is the reversed `ring` condition, not merely the choice of NOT operator.

Entry 85 - Q5b serial two's complementer, true one-hot Mealy FSM

The problem asks for one-hot encoding. The latest successful code uses `s0=2'b00` and `s1=2'b01`; `2'b00` has no asserted state bit, so it is not one-hot even though HDLBits accepts the external behavior. Functional tests cannot necessarily observe the internal encoding.

The older unsuccessful submission also omitted `zr=0` in one branch of its combinational block. That leaves `zr` unassigned on that path and infers a latch. Here is a one-hot implementation with every combinational output assigned:

```
localparam [1:0] BEFORE_FIRST_ONE = 2'b01;
localparam [1:0] AFTER_FIRST_ONE  = 2'b10;

reg [1:0] state, next_state;

always @(posedge clk or posedge areset) begin
    if (areset)
        state <= BEFORE_FIRST_ONE;
    else
        state <= next_state;
end

always @(*) begin
    next_state = state;
    z = 1'b0;

    case (1'b1)
        state[0]: begin
            z = x;                  // Copy zeros and the first one.
            if (x)
                next_state = AFTER_FIRST_ONE;
        end
        state[1]: begin
            z = ~x;                 // Invert all later bits.
            next_state = AFTER_FIRST_ONE;
        end
        default: begin
            next_state = BEFORE_FIRST_ONE;
            z = x;
        end
    endcase
end
```

The 3 September follow-up: entries 86-93

Kapil reported these eight problems completed in the current pass. The Chrome tab strip captured on 2026-09-03 contains the same eight HDLBits pages, in the same progress block. Because page-body attachment was unavailable during this follow-up, the evidence column records the confirmation without adding submission timestamps.

Entry	Problem	Attempt 2 state	Question/reference
86	Vectorr	Done	No questions asked
87	Dualedge	Done	Day 6 non-FSM Q&A PDF
88	Thermostat	Done	No questions asked
89	Sim/circuit5	Done	Day 6 non-FSM Q&A PDF
90	Exams/2014 q3fsm	Done	Standalone three-sample-window FSM PDF
91	Vector4	Done	No questions asked
92	Count15	Done	No questions asked
93	Popcount3	Done	No questions asked

The saved Edge conversations add three review topics:

- In Dualedge, `qpos | qneg` can preserve a stale 1 from the register that was not updated at the latest edge. The primary HDLBits solution selects `qpos` while `clk` is high and `qneg` while it is low. The expanded PDF now distinguishes simulator acceptance from FPGA implementability: ordinary fabric registers are normally single-edge, whereas real DDR capture and launch use device-specific I/O resources such as AMD IDDR/ODDR or Intel/Altera DDIO. It also covers half-cycle timing, duty-cycle, glitch, metastability, reset, and XOR-startup concerns using primary vendor documentation.

- In Sim/circuit5, the waveform implies `c=0 -> b`, `c=1 -> e`, `c=2 -> a`, `c=3 -> d`, and all other selector values -> `4'hf`. Therefore `c` belongs in `case(c)`; input bus `b` is an identifier, not hexadecimal digit B.

- In Exams/2014 q3fsm, `ticks==2` means two samples have already been processed before the current third edge. Because nonblocking assignments expose old register values within the block, the third-sample decision uses `count + w`. The window counters must reset after every third sample, not only when the group contains exactly two ones. Both earlier wrong versions and the corrected two-state RTL are preserved in the standalone PDF.

Local verification

The reusable testbench day5_original_submission_selfcheck.sv checks:

- all 16 input combinations for entry 77;

- sampled 0-to-1 pulses for entry 74;

- overlapping `10101` recognition and immediate reset assertion for entry 75;

- LSB-first Moore and true one-hot Mealy two's-complement streams for entries 80 and 85;

- all 16 waveform input combinations for entry 81;

- 200 randomized concatenations for entry 82;

- sticky falling-edge capture, reset priority, and post-reset behavior for entry 83;

- all four Ringer truth-table rows and mutual exclusion for entry 84.

Icarus Verilog result:

```
PASS: Chrome-audit follow-up behavior checks passed (entries 74, 75, 77, 80-85).
```

The Day 6 checkers add:

```
PASS: all 8 possible q3fsm groups plus back-to-back grouping; saved reset-only-on-match
  bug diverged as expected
PASS: Dualedge MUX/XOR solutions and all 16 Circuit5 selector values; stale-register OR
  bug diverged as expected
```

Historical submission records

The table below records every original page in the audited range. For entries 1-85, dates are copied from the saved-submission selector displayed by HDLBits. For entries 86-93, `2026-09-03 current pass; user-confirmed` records Kapil's confirmation rather than a submission time.

No.	Original HDLBits page	Last success shown in Chrome	Last non-success shown in Chrome	Attempt 2 state	Question/reference
1	Lemmings1	8/29/2026, 3:33:16 PM	8/29/2026, 3:30:28 PM	Done	No explicit question recorded
2	Fsm serial	8/30/2026, 9:56:33 AM	8/30/2026, 9:55:55 AM	Done	Serial discussion PDF
3	Lemmings2	8/29/2026, 3:54:58 PM	8/29/2026, 3:51:51 PM	Done	No explicit question recorded
4	Fsm serialdata	8/30/2026, 10:24:16 AM	8/30/2026, 10:22:16 AM	Done	Serial discussion PDF
5	Lemmings3	8/29/2026, 4:21:22 PM	8/29/2026, 4:21:03 PM	Done	No explicit question recorded
6	Fsm serialdp	8/30/2026, 10:55:13 AM	8/30/2026, 10:53:43 AM	Done	Serial discussion PDF
7	Lemmings4	8/29/2026, 9:14:52 PM	8/29/2026, 9:12:30 PM	Done	Combined questions PDF
8	Fsm hdlc	8/29/2026, 11:20:34 PM	8/29/2026, 11:14:37 PM	Done	HDLC discussion PDF
9	Conditional	8/30/2026, 3:40:27 PM	none	Done	No explicit question recorded
10	Dff	8/30/2026, 3:42:19 PM	none	Done	No explicit question recorded
11	Step one	8/30/2026, 3:42:48 PM	none	Done	No explicit question recorded
12	Fsm1	8/30/2026, 3:49:25 PM	8/30/2026, 3:48:09 PM	Done	No explicit question recorded
13	Bugs mux2	8/30/2026, 4:09:04 PM	8/30/2026, 4:08:09 PM	Done	Combined questions PDF
14	Reduction	8/30/2026, 4:10:13 PM	6/24/2026, 6:14:19 PM	Done	No explicit question recorded
15	Dff8	8/30/2026, 4:12:23 PM	none	Done	No explicit question recorded
16	Zero	8/30/2026, 4:12:48 PM	none	Done	No explicit question recorded
17	Fsm1s	8/30/2026, 4:21:45 PM	8/30/2026, 4:23:11 PM	Done	No explicit question recorded
18	Gates100	8/30/2026, 4:14:01 PM	none	Done	No explicit question recorded
19	Dff8r	8/30/2026, 5:06:27 PM	6/25/2026, 10:23:54 PM	Done	Combined questions PDF
20	Wire	8/30/2026, 5:06:45 PM	none	Done	Combined questions PDF
21	Bugs nand3	8/30/2026, 5:12:57 PM	8/30/2026, 5:07:56 PM	Done	Combined questions PDF
22	Fsm2	8/30/2026, 5:15:53 PM	8/30/2026, 5:15:28 PM	Done	Combined questions PDF
23	Vector100r	8/30/2026, 5:21:27 PM	6/24/2026, 6:21:32 PM	Done	Combined questions PDF
24	Dff8p	8/30/2026, 6:12:50 PM	8/30/2026, 5:30:37 PM	Done	Combined questions PDF
25	Wire4	8/30/2026, 5:31:49 PM	6/23/2026, 5:49:49 PM	Done	Combined questions PDF
26	Popcount255	8/30/2026, 5:50:16 PM	8/30/2026, 5:46:15 PM	Done	Combined questions PDF
27	Fsm2s	8/30/2026, 5:54:41 PM	8/30/2026, 5:53:48 PM	Done	Combined questions PDF
28	Dff8ar	8/30/2026, 5:58:16 PM	6/25/2026, 10:28:54 PM	Done	Combined questions PDF
29	Bugs mux4	8/30/2026, 6:09:06 PM	8/30/2026, 6:08:27 PM	Done	Combined questions PDF
30	Adder100i	8/30/2026, 6:29:42 PM	8/30/2026, 6:29:17 PM	Done	Combined questions PDF

No.	Original HDLBits page	Last success shown in Chrome	Last non-success shown in Chrome	Attempt 2 state	Question/reference
31	Notgate	8/30/2026, 7:09:37 PM	none	Done	Combined questions PDF
32	Fsm3comb	8/30/2026, 11:45:51 PM	8/30/2026, 11:43:15 PM	Done	No explicit question recorded
33	Dff16e	8/30/2026, 7:26:48 PM	6/25/2026, 10:39:59 PM	Done	No explicit question recorded
34	Bcdadd100	8/30/2026, 9:57:17 PM	8/30/2026, 9:54:23 PM	Done	No explicit question recorded
35	Andgate	8/30/2026, 7:11:50 PM	none	Done	No explicit question recorded
36	Bugs addsubz	8/30/2026, 10:07:09 PM	8/30/2026, 10:02:03 PM	Done	Combined questions PDF
37	Exams/m2014 q4h	8/30/2026, 7:21:33 PM	none	Done	No explicit question recorded
38	Exams/m2014 q4a	8/30/2026, 7:32:02 PM	8/30/2026, 7:30:55 PM	Done	Combined questions PDF
39	Fsm3onehot	8/30/2026, 10:36:07 PM	8/30/2026, 10:35:24 PM	Done	No explicit question recorded
40	Norgate	8/30/2026, 7:22:24 PM	6/24/2026, 12:31:16 AM	Done	No explicit question recorded
41	Exams/m2014 q4i	8/30/2026, 7:22:37 PM	none	Done	No explicit question recorded
42	Exams/m2014 q4b	8/30/2026, 7:37:09 PM	none	Done	No explicit question recorded
43	Fsm3	8/30/2026, 10:57:04 PM	none	Done	No explicit question recorded
44	Xnorgate	8/30/2026, 7:38:46 PM	8/30/2026, 7:38:25 PM	Done	No explicit question recorded
45	Exams/m2014 q4e	8/30/2026, 7:33:30 PM	6/24/2026, 11:32:45 PM	Done	No explicit question recorded
46	Exams/m2014 q4c	8/30/2026, 10:40:45 PM	none	Done	No explicit question recorded
47	Always case	8/30/2026, 10:43:52 PM	8/30/2026, 10:43:30 PM	Done	No explicit question recorded
48	Fsm3s	8/30/2026, 10:54:27 PM	8/30/2026, 10:52:35 PM	Done	No explicit question recorded
49	Wire decl	8/30/2026, 10:46:25 PM	none	Done	No explicit question recorded
50	Exams/m2014 q4f	8/30/2026, 7:35:50 PM	8/30/2026, 7:35:13 PM	Done	No explicit question recorded
51	Exams/m2014 q4d	8/31/2026, 9:32:17 AM	8/31/2026, 9:31:28 AM	Done	No explicit question recorded
52	Exams/m2014 q4g	8/31/2026, 9:34:10 AM	none	Done	No explicit question recorded
53	7458	8/31/2026, 9:39:14 AM	none	Done	No explicit question recorded
54	Exams/ece241 2013 q4	9/1/2026, 12:21:28 AM	9/1/2026, 12:16:46 AM	Done	Recovered questions PDF
55	Sim/circuit1	8/31/2026, 8:36:12 PM	6/28/2026, 1:32:14 AM	Done	No explicit question recorded
56	Mt2015 muxdff	9/1/2026, 12:32:44 AM	9/1/2026, 12:32:23 AM	Done	No explicit question recorded
57	Gates	9/1/2026, 12:39:20 AM	9/1/2026, 12:38:40 AM	Done	No explicit question recorded
58	Vector0	9/1/2026, 12:43:49 AM	9/1/2026, 12:40:57 AM	Done	No explicit question recorded
59	Fsm onehot	9/1/2026, 12:58:13 AM	7/5/2026, 5:07:43 PM	Done	No explicit question recorded
60	Exams/2014 q4a	9/1/2026, 1:04:13 AM	6/26/2026, 12:19:52 AM	Done	No explicit question recorded
61	7420	9/1/2026, 1:28:08 AM	none	Done	No explicit question recorded
62	Vector1	9/1/2026, 1:30:34 AM	6/23/2026, 9:44:41 PM	Done	No explicit question recorded
63	Sim/circuit2	9/1/2026, 11:00:02 AM	9/1/2026, 10:53:57 AM	Done	No explicit question recorded

No.	Original HDLBits page	Last success shown in Chrome	Last non-success shown in Chrome	Attempt 2 state	Question/reference
64	Fsm ps2	9/1/2026, 4:25:35 PM	9/1/2026, 4:26:58 PM	Done	No explicit question recorded
65	Truthtable1	9/1/2026, 5:12:37 PM	none	Done	No explicit question recorded
66	Exams/ece241 2014 q4	9/1/2026, 5:28:19 PM	9/1/2026, 5:26:37 PM	Done	Day 4 questions PDF
67	Vector2	9/1/2026, 5:40:40 PM	9/1/2026, 5:38:14 PM	Done	Day 4 questions PDF
68	Mt2015 eq2	9/1/2026, 6:30:47 PM	none	Done	No explicit question recorded
69	Exams/ece241 2013 q7	9/1/2026, 6:29:32 PM	none	Done	No explicit question recorded
70	Fsm ps2data	9/1/2026, 4:46:09 PM	9/1/2026, 4:43:36 PM	Done	Day 4 questions PDF
71	Sim/circuit3	9/1/2026, 10:21:36 PM	9/1/2026, 10:17:52 PM	Done	No explicit question recorded
72	Mt2015 q4a	9/1/2026, 10:22:14 PM	none	Done	No explicit question recorded
73	Vectorgates	9/1/2026, 10:24:57 PM	9/1/2026, 10:23:57 PM	Done	No explicit question recorded
74	Edgedetect	9/1/2026, 10:28:38 PM	9/1/2026, 9:58:30 PM	Done	Recovered questions PDF
75	Exams/ece241 2013 q8	9/1/2026, 10:37:57 PM	9/1/2026, 10:36:46 PM	Done	Recovered questions PDF
76	Mt2015 q4b	9/1/2026, 10:38:41 PM	none	Done	No explicit question recorded
77	Gates4	9/1/2026, 10:44:37 PM	6/23/2026, 10:15:40 PM	Done	Recovered questions PDF
78	Edgedetect2	9/1/2026, 10:47:42 PM	9/1/2026, 10:47:07 PM	Done	No explicit question recorded
79	Mt2015 q4	9/1/2026, 11:01:04 PM	9/1/2026, 11:00:16 PM	Done	No explicit question recorded
80	Exams/ece241 2014 q5a	9/2/2026, 4:28:11 PM	9/2/2026, 4:18:12 PM	Done	Entry 80 Moore FSM PDF
81	Sim/circuit4	9/1/2026, 11:26:08 PM	6/28/2026, 3:34:12 PM	Done	Day 5 submission review
82	Vector3	9/1/2026, 11:23:20 PM	9/1/2026, 11:19:13 PM	Done	Day 5 submission review
83	Edgecapture	9/2/2026, 5:00:08 PM	9/2/2026, 4:57:13 PM	Done	Day 5 submission review
84	Ringer	9/2/2026, 4:54:21 PM	9/2/2026, 4:53:47 PM	Done	Day 5 submission review
85	Exams/ece241 2014 q5b	9/2/2026, 4:44:54 PM	9/2/2026, 4:31:02 PM	Done	Day 5 submission review
86	Vectorr	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	No explicit question recorded
87	Dualedge	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	Day 6 non-FSM Q&A PDF
88	Thermostat	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	No explicit question recorded
89	Sim/circuit5	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	Day 6 non-FSM Q&A PDF
90	Exams/2014 q3fsm	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	Standalone three-sample-window FSM PDF
91	Vector4	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	No explicit question recorded

No.	Original HDLBits page	Last success shown in Chrome	Last non-success shown in Chrome	Attempt 2 state	Question/reference
92	Count15	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	No explicit question recorded
93	Popcount3	2026-09-03 current pass; user-confirmed	not captured in follow-up	Done	No explicit question recorded

At the 3 September checkpoint, entries 1-93 were Done and entries 94-178 had not yet been included in the review. The earlier 7 September document-review checkpoint recorded 151 Done and 27 Pending. These are historical counts; use the tracker for current progress.